

AJINKYA WADASKAR

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SUMMARY

AI / ML Engineer and Data Scientist with 6+ years building production LLM applications, agentic systems, and ML pipelines. Engineered LangChain and GPT-4 agents (~80% effort reduction), an LLM evaluation harness, and a hybrid AI and rule-based debug agent — with rigorous LLM evaluation (DeepEval) and guardrails. Production ML (segmentation, 11.2% retention lift) on a strong experimentation / causal-inference foundation (4% conversion lift). Python, SQL, GCP.

WORK EXPERIENCE

Ford Motor Company — Dearborn, MI

05/2022 – Present

Data Scientist / AI Engineer

- Cut ETL migration effort ~80% by engineering a LangChain and GPT-4 agentic LLM pipeline that auto-translates legacy Alteryx workflows into production BigQuery SQL and Dataform DAGs, accelerating a multi-year modernization backlog.
- Built an LLM evaluation harness measuring SQL-translation accuracy and schema-match recall against complex source workflows, catching regressions before production cutover and reducing manual QA.
- Built a hybrid AI and rule-based debug agent (Python, BigQuery, Claude API) for the financial-close tie-out that automates anomaly detection, reconciliation, and root-cause diagnosis (13 detection checks) — cutting per-issue triage from hours to minutes and saving 1–2 days per monthly close.
- Hardened analytics reliability with CI/CD-driven Dataform jobs, scheduled backfills, lineage documentation, and data-quality tests (schema, not-null, uniqueness) across JSON/CSV/Parquet pipelines spanning billing, geospatial, and telematics domains.
- Built and deployed production ML user-segmentation models (scikit-learn K-Means, behavioral feature engineering) separating power from casual users, with targeted lifecycle re-engagement that lifted retention 11.2%.
- Lifted conversion rate 4% by owning end-to-end experiment design — hypothesis framing, power analysis, difference-in-differences (diff-in-diff) causal inference, CUPED variance reduction, and guardrail metrics — partnering with Product and Marketing on funnel A/B experiments with anomaly detection.

General Motors — Warren, MI

04/2021 – 05/2022

Data Analyst

- Built Python and SQL data pipelines over large-scale operational and telematics data (Hadoop, Azure, SQL Server) and Power BI data models that improved data integrity and accelerated cross-functional product and operations decisions.

Peterson American Corporation — Southfield, MI

08/2019 – 04/2021

Data Analyst / Data Engineer

- Saved 10 hours/week and achieved 40% faster query performance by automating Python and Power Automate data pipelines across SAP ERP, SQL Server, and Db2, and tuning SQL for Finance and Operations stakeholders.

Grupo Antolin North America — Auburn Hills, MI

08/2018 – 04/2019

Data Analyst, Warranty

- Boosted reporting efficiency 70% by automating ETL and Power BI models on OEM warranty cost-claim data, applying data-mining to surface high-cost failure modes for finance and quality teams.

PROJECTS

Credit Decision Explainer Agent — github.com/ajinkyawadaskar/credit-decision-explainer

XGBoost probability-of-default credit-risk model with SHAP explainability and an FCRA-compliant LLM that drafts adverse-action notices; LangGraph and FastAPI, evaluated in CI with DeepEval.

Data Analyst Agent — github.com/ajinkyawadaskar/data-analyst-agent

LangGraph agent converting natural-language questions into validated BigQuery SQL via Claude / GPT-4 tool-calling, with a DeepEval suite scoring answer accuracy.

SKILLS

AI & LLMs: Large Language Models (LLMs), LangChain, LangGraph, GPT-4 / Claude API, Claude Code, Agentic Workflows, Tool-Calling / Function Calling, Multi-Step Reasoning, Schema-Grounded Retrieval, Prompt Engineering, Guardrails, Natural-Language-to-SQL, MCP

Machine Learning: Feature Engineering, Model Evaluation, Logistic Regression, XGBoost, LightGBM, K-Means Clustering, Time-Series Forecasting, Churn / LTV / Propensity Modeling, Credit-Risk Modeling, Model Explainability (SHAP)

MLOps & Data Engineering: Python (Pandas, NumPy, scikit-learn, statsmodels), SQL, PySpark, Spark, Git; GCP (BigQuery, BigQuery ML, Dataform, Dataflow, Vertex AI), Azure, Docker, CI/CD, MLflow, Airflow, dbt, Hadoop, ETL/ELT, Production ML / Model Deployment & Monitoring

LLM Evaluation & Observability: LLM Evaluation (DeepEval), Regression Testing, Accuracy / Recall Metrics, Hallucination & Failure-Mode Analysis, Audit Trails, Model Monitoring, AI-Assisted Reporting

Languages & Tools: Python (Pandas, NumPy, scikit-learn, statsmodels), SQL, PySpark, Git / GitHub, FastAPI / REST APIs, Docker; Tableau, Amplitude, Power BI, Qlik Sense

EDUCATION

M.S., Industrial Engineering — Wayne State University, Detroit, MI

08/2017 – 05/2019

B.E., Mechanical Engineering — Amravati University, India

07/2011 – 06/2015